

Feb 24, 2024

Total

25.5/27

Knowledge

1. Divide $6x^3 + 20x^2 + 5$ by $3x + 1$ using long division. Write the corresponding division statement and state restrictions on x .

(3)

$$\begin{array}{r}
 2x^2 + 6x - 2 \\
 3x+1 \overline{) 6x^3 + 20x^2 + 0x + 5} \\
 \underline{-(6x^3 + 2x^2)} \\
 18x^2 + 0x \\
 \underline{-(18x^2 + 6x)} \\
 -6x + 5 \\
 \underline{-(-6x - 2)} \\
 7R
 \end{array}$$

$$6x^3 + 20x^2 + 5 = (3x+1)(2x^2 + 6x - 2) + 7$$

$$= 2(3x+1)(x^2 + 3x - 1) + 7$$

$$x \neq -\frac{1}{3}$$

no need,
you're not
factoring.

2. Determine the remainder when $3x^3 + 2x^2 - 3x - 2$ is divided by $2x - 5$.

(2)

$$3\left(\frac{5}{2}\right)^3 + 2\left(\frac{5}{2}\right)^2 - 3\left(\frac{5}{2}\right) - 2 = \frac{399}{8} R$$

3. Factor.

a) $4y^2 + 4yz + z^2 - 1$

(1)

$$\begin{aligned}
 &= (2y+z)^2 - 1^2 \\
 &= (2y+z+1)(2y+z-1)
 \end{aligned}$$

b) $3(x-1)^2 - 20(x-1) - 7$

(2)

$$\begin{aligned}
 &= [3(x-1)+1][(x-1)-7] \\
 &= (3x-2)(x-8)
 \end{aligned}$$

c) $x^3 - (x+3)^3$

$$= [x - (x+3)][x^2 + x(x+3) + (x+3)^2]$$

(3)

$$= -3(x^2 + x^2 + 3x + x^2 + 6x + 9)$$

$$= -3(3x^2 + 9x + 9)$$

$$= -9(x^2 + 3x + 3)$$

d) $18x^3 + 3x^2 - 1 - 4x$

$$= 18x^3 + 3x^2 - 4x - 1$$

(2)
(3)

$$\begin{array}{r}
 \frac{1}{2} \cdot \begin{array}{rrrr} 18 & 3 & -4 & -1 \\ & \downarrow & 9 & 6 & 1 \\ \hline 18 & 12 & 2 & 0R \end{array}
 \end{array}$$

$$= (x - \frac{1}{2})(18x^2 + 12x + 2)$$

$$= (2x-1)(9x^2 + 6x + 1)$$

↓

$\pm 1, 2, 3, 5, 6, 10, 15, 30$

e) $x^4 + x^3 - 31x^2 - 61x - 30$

$$\begin{array}{r|rrrrr} -1 & 1 & 1 & -31 & -61 & -30 \\ & \downarrow & -1 & 0 & 31 & 30 \\ \hline -1 & 1 & 0 & -31 & -30 & 0R \\ & \downarrow & -1 & 1 & 30 & \\ \hline & 1 & -1 & -30 & 0R & \end{array}$$

$= (x+1)^2(x+5)(x-6)$

f) $216x^3 + 2197$

$= (6x)^3 + (13)^3$

$= (6x+13)(36x^2-78x+169)$

4. Given the diagram below: Complete the partially completed synthetic division, where the divisor is $x+3$ and fill in the blank below.

$$\begin{array}{r|rrrr} -3 & a & -4 & 3 & 101 \\ & \downarrow & -9 & 39 & -126 \\ \hline & 3 & b & c & -25 \\ & & -13 & 42 & \end{array}$$

$a=3$
 $b=-13$
 $c=42$
 $d=101$

$a+b+c+d = 133$

5. a) A cylindrical can, $(x+2)$ cm high, has a volume of $(4\pi x^3 + 28\pi x^2 + 65\pi x + 50\pi)$ cm³. What is the radius of the can?

$\pi(4x^3 + 28x^2 + 65x + 50)$
 $\pm 1, 2, 4 \quad \pm 1, 2, 5, 10, 25, 50$

$$\begin{array}{r|rrrr} -2 & 4 & 28 & 65 & 50 \\ & \downarrow & -8 & -40 & -50 \\ \hline & 4 & 20 & 25 & 0R \end{array}$$

$V = \pi r^2 h$
 $= \pi(2x+5)^2(x+2)$
 $r = (2x+5)$ cm

- b) What is the restriction on x ?

$x \neq -2$

6. For what value of b will the polynomial $P(x) = -3x^3 - bx^2 + 3x + 2$ have the same remainder when it is divided by $x+1$ and $2x-1$?

$P(-1) = -3(-1)^3 - b(-1)^2 + 3(-1) + 2$

$R = 3 - b - 3 + 2$

$0R = 2 - b$

$P(\frac{1}{2}) = -3(\frac{1}{2})^3 - b(\frac{1}{2})^2 + 3(\frac{1}{2}) + 2$

$R = -\frac{3}{8} - \frac{b}{4} + \frac{3}{2} + 2$

$R = \frac{25}{8} - \frac{b}{4}$

$4 \times ① - ②: 4R - 4R = (8 - \frac{25}{2}) + (-4b - (-b))$

$0 = -\frac{9}{2} - 3b$

$b = -\frac{3}{2}$

$b = -\frac{3}{2}$

$② 4R = \frac{25}{2} - b$